



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)
FULL SYLLABUS
28-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

Time : 3 Hours

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2022

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PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

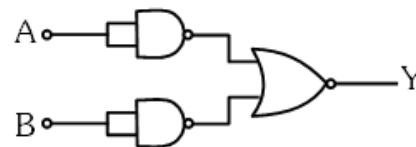
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- For a gas $C_p - C_v = R$ in a state P and $C_p - C_v = 1.10 R$ in a state Q, T_P and T_Q are the temperatures in two different states P and Q respectively. Then
 (A) $T_P = T_Q$ (B) $T_P < T_Q$
 (C) $T_P = 0.9 T_Q$ (D) $T_P > T_Q$
- Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A : Moment of inertia of a circular disc of mass 'M' and radius 'R' about X, Y axes (passing through its plane) and Z-axis which is perpendicular to its plane were found to be I_x , I_y and I_z respectively. The respective radii of gyration about all the three axes will be the same.
Reason R : A rigid body making rotational motion has fixed mass and shape. In the light of the above statements, choose the most appropriate answer from the options given below :
 (A) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.
 (B) **A** is not correct but **R** is correct.
 (C) **A** is correct but **R** is not correct.
 (D) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

- What should be the order of arrangement of de-Broglie wavelength of electron (λ_e), an α -particle (λ_α) and proton (λ_p) given that all have the same kinetic energy ?
 (A) $\lambda_e = \lambda_p = \lambda_\alpha$ (B) $\lambda_e < \lambda_p < \lambda_\alpha$
 (C) $\lambda_e > \lambda_p > \lambda_\alpha$ (D) $\lambda_e = \lambda_p > \lambda_\alpha$

- Identify the logic operation carried out.



- (A) OR (B) AND
 (C) NOR (D) NAND
- A particle of mass $4M$ at rest disintegrates into two particles of mass M and $3M$ respectively having non zero velocities. The ratio of de-Broglie wavelength of particle of mass M to that of mass $3M$ will be :
 (A) $1 : 3$ (B) $3 : 1$
 (C) $1 : \sqrt{3}$ (D) $1 : 1$
 - Some nuclei of a radioactive material are undergoing radioactive decay. The time gap between the instances when a quarter of the nuclei have decayed and when half of the nuclei have decayed is given as : (where λ is the decay constant)
 (A) $\frac{1}{2} \frac{\ln 2}{\lambda}$ (B) $\frac{\ln 2}{\lambda}$
 (C) $\frac{2 \ln 2}{\lambda}$ (D) $\frac{\ln \frac{3}{2}}{\lambda}$

7. Match List I with List II.

	List-I		List-II
(a)	$\vec{C} - \vec{A} - \vec{B} = 0$	(i)	
(b)	$\vec{A} - \vec{C} - \vec{B} = 0$	(ii)	
(c)	$\vec{B} - \vec{A} - \vec{C} = 0$	(iii)	
(d)	$\vec{A} + \vec{B} = -\vec{C}$	(iv)	

Choose the correct answer from the options given below :

- (A) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (ii)
 (B) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (i), (d) $\rightarrow$ (ii)
 (C) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)
 (D) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (iii)

8. A parallel plate capacitor with plate area 'A' and distance of separation 'd' is filled with a dielectric. What is the capacity of the capacitor when permittivity of the dielectric varies as :

$$\epsilon(x) = \epsilon_0 + kx, \text{ for } \left(0 < x \leq \frac{d}{2}\right)$$

$$\epsilon(x) = \epsilon_0 + k(d - x), \text{ for } \left(\frac{d}{2} \leq x \leq d\right)$$

- (A) $\left(\epsilon_0 + \frac{kd}{2}\right)^{2/kA}$ (B) $\frac{kA}{2 \ln\left(\frac{2\epsilon_0 + kd}{2\epsilon_0}\right)}$
 (C) 0 (D) $\frac{kA}{2} \ln\left(\frac{2\epsilon_0}{2\epsilon_0 - kd}\right)$

9. A monoatomic ideal gas, initially at temperature T_1 is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature T_2 by releasing the piston suddenly. If ℓ_1 and ℓ_2 are the lengths of the gas column, before and after the expansion respectively, then the value of $\frac{T_1}{T_2}$ will be :

- (A) $\left(\frac{\ell_1}{\ell_2}\right)^{\frac{2}{3}}$ (B) $\left(\frac{\ell_2}{\ell_1}\right)^{\frac{2}{3}}$
 (C) $\frac{\ell_2}{\ell_1}$ (D) $\frac{\ell_1}{\ell_2}$

10. A ray of laser of a wavelength 630 nm is incident at an angle of 30° at the diamond-air interface. It is going from diamond to air. The refractive index of diamond is 2.42 and that of air is 1. Choose the correct option.

- (A) angle of refraction is 24.41°
 (B) angle of refraction is 30°
 (C) refraction is not possible
 (D) angle of refraction is 53.4°

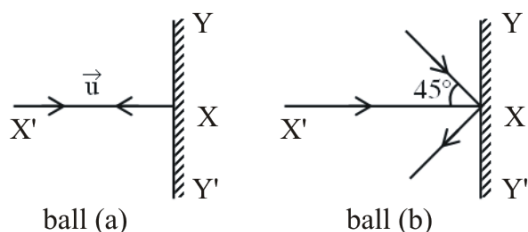
11. Two wires of same length and radius are joined end to end and loaded. The Young's moduli of the materials of the two wires are Y_1 and Y_2 . The combination behaves as a single wire then its Young's modulus is :

- (A) $Y = \frac{2Y_1Y_2}{3(Y_1 + Y_2)}$
 (B) $Y = \frac{2Y_1Y_2}{Y_1 + Y_2}$
 (C) $Y = \frac{Y_1Y_2}{2(Y_1 + Y_2)}$
 (D) $Y = \frac{Y_1Y_2}{Y_1 + Y_2}$

12. The half-life of ^{198}Au is 3 days. If atomic weight of ^{198}Au is 198 g/mol then the activity of 2 mg of ^{198}Au is [in disintegration/second] :

(A) 2.67×10^{12}
 (B) 6.06×10^{18}
 (C) 32.36×10^{12}
 (D) 16.18×10^{12}

13. Two billiard balls of equal mass 30 g strike a rigid wall with same speed of 108 kmph (as shown) but at different angles. If the balls get reflected with the same speed then the ratio of the magnitude of impulses imparted to ball 'a' and ball 'b' by the wall along 'X' direction is :

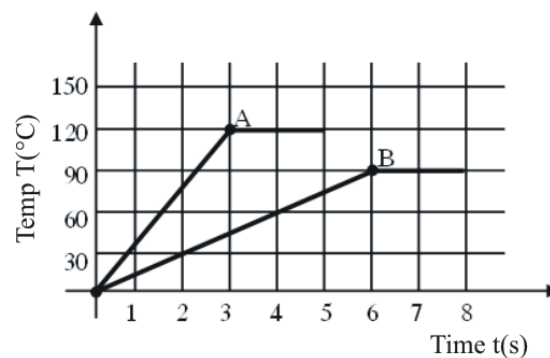


(A) 1 : 1
 (B) $\sqrt{2} : 1$
 (C) 2 : 1
 (D) $1 : \sqrt{2}$

14. In the Young's double slit experiment, the distance between the slits varies in time as $d(t) = d_0 + a_0 \sin \omega t$; where d_0 , ω and a_0 are constants. The difference between the largest fringe width and the smallest fringe width obtained over time is given as :

(A) $\frac{2\lambda D(d_0)}{(d_0^2 - a_0^2)}$
 (B) $\frac{2\lambda D a_0}{(d_0^2 - a_0^2)}$
 (C) $\frac{\lambda D}{d_0^2} a_0$
 (D) $\frac{\lambda D}{d_0 + a_0}$

15. Two different metal bodies A and B of equal mass are heated at a uniform rate under similar conditions. The variation of temperature of the bodies is graphically represented as shown in the figure. The ratio of specific heat capacities is :



(A) $\frac{8}{3}$
 (B) $\frac{3}{8}$
 (C) $\frac{3}{4}$
 (D) $\frac{4}{3}$

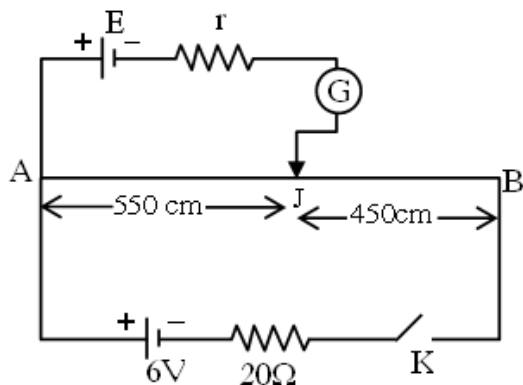
16. A linearly polarized electromagnetic wave in vacuum is

$$E = 3.1 \cos [(1.8)z - (5.4 \times 10^6)t] \hat{i} \text{ N/C}$$

is incident normally on a perfectly reflecting wall at $z = a$. Choose the correct option

(A) The wavelength is 5.4 m
 (B) The frequency of electromagnetic wave is 54×10^4 Hz.
 (C) The transmitted wave will be $3.1 \cos [(1.8)z - (5.4 \times 10^6)t] \hat{i} \text{ N/C}$
 (D) The reflected wave will be $3.1 \cos [(1.8)z + (5.4 \times 10^6)t] \hat{i} \text{ N/C}$

17. In the given figure, there is a circuit of potentiometer of length $AB = 10 \text{ m}$. The resistance per unit length is $0.1 \Omega \text{ per cm}$. Across AB , a battery of emf E and internal resistance ' r ' is connected. The maximum value of emf measured by this potentiometer is :



- (A) 5 V
(B) 2.25 V
(C) 6 V
(D) 2.75 V
18. In amplitude modulation, the message signal $V_m(t) = 10 \sin(2\pi \times 10^5 t)$ volts and Carrier signal $V_c(t) = 20 \sin(2\pi \times 10^7 t)$ volts. The modulated signal now contains the message signal with lower side band and upper side band frequency, therefore the bandwidth of modulated signal is $\alpha \text{ kHz}$. The value of α is :
- (A) 200 kHz
(B) 50 kHz
(C) 100 kHz
(D) 0

19. Water droplets are coming from an open tap at a particular rate. The spacing between a droplet observed at 4th second after its fall to the next droplet is 34.3 m. At what rate the droplets are coming from the tap ? (Take $g = 9.8 \text{ m/s}^2$)

- (A) 3 drops / 2 seconds
(B) 2 drops / second
(C) 1 drop / second
(D) 1 drop / 7 seconds
20. The minimum and maximum distances of a planet revolving around the Sun are x_1 and x_2 . If the minimum speed of the planet on its trajectory is v_0 then its maximum speed will be :

- (A) $\frac{v_0 x_1^2}{x_2^2}$
(B) $\frac{v_0 x_2^2}{x_1^2}$
(C) $\frac{v_0 x_1}{x_2}$
(D) $\frac{v_0 x_2}{x_1}$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

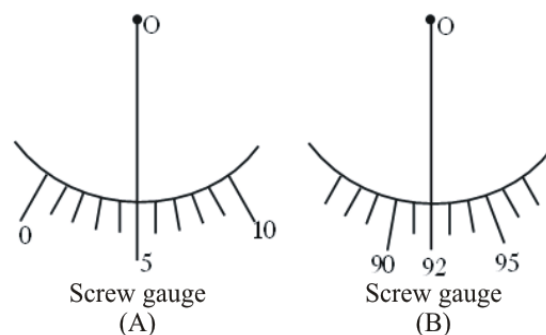
Negative Marks : -1 If wrong option is selected.

1. A body of mass 2 kg moving with a speed of 4 m/s. makes an elastic collision with another body at rest and continues to move in the original direction but with one fourth of its initial speed. The speed of the two body centre of mass is $\frac{x}{10}$ m/s. Then the value of x is _____.

2. Student A and Student B used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is 0.322 cm. The absolute value of the difference between the final circular scale readings observed by the students A and B is _____.

[Figure shows position of reference 'O' when jaws of screw gauge are closed]

Given pitch = 0.1 cm.



3. An inductor of 10 mH is connected to a 20 V battery through a resistor of 10 k Ω and a switch. After a long time, when maximum current is set up in the circuit, the current is switched off. The current in the circuit after 1 μ s is $\frac{x}{100}$ mA. Then x is equal to _____. (Take $e^{-1} = 0.37$)

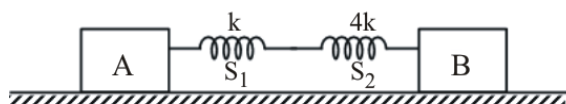
4. A circular conducting coil of radius 1 m is being heated by the change of magnetic field $\vec{B}$ passing perpendicular to the plane in which the coil is laid. The resistance of the coil is $2 \mu\Omega$. The magnetic field is slowly switched off such that its magnitude changes in time as

$$B = \frac{4}{\pi} \times 10^{-3} \text{ T} \left(1 - \frac{t}{100} \right)$$

The energy dissipated by the coil before the magnetic field is switched off completely is

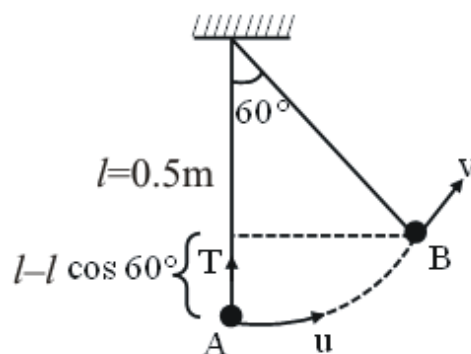
$$E = \text{_____ mJ.}$$

5. In the reported figure, two bodies A and B of masses 200 g and 800 g are attached with the system of springs. Springs are kept in a stretched position with some extension when the system is released. The horizontal surface is assumed to be frictionless. The angular frequency will be _____ rad/s when $k = 20 \text{ N/m}$.



6. The value of aluminium susceptibility is 2.2×10^{-5} . The percentage increase in the magnetic field if space within a current carrying toroid is filled with aluminium is $\frac{x}{10^4}$. Then the value of x is _____.

7. A particle of mass 1 mg and charge q is lying at the mid-point of two stationary particles kept at a distance '2 m' when each is carrying same charge 'q'. If the free charged particle is displaced from its equilibrium position through distance 'x' ($x \ll 1 \text{ m}$). The particle executes SHM. Its angular frequency of oscillation will be _____ $\times 10^5 \text{ rad/s}$ if $q^2 = 10 \text{ C}^2$.
8. An electric bulb rated as 200 W at 100 V is used in a circuit having 200 V supply. The resistance 'R' that must be put in series with the bulb so that the bulb delivers the same power is _____ Ω .
9. A pendulum bob has a speed of 3 m/s at its lowest position. The pendulum is 50 cm long. The speed of bob, when the length makes an angle of 60° to the vertical will be ($g = 10 \text{ m/s}^2$) _____ m/s.



10. A particle of mass 'm' is moving in time 't' on a trajectory given by $\vec{r} = 10 \alpha t^2 \hat{i} + 5\beta (t - 5)\hat{j}$ Where α and β are dimensional constants. The angular momentum of the particle becomes the same as it was for $t = 0$ at time $t = \text{_____}$ seconds.

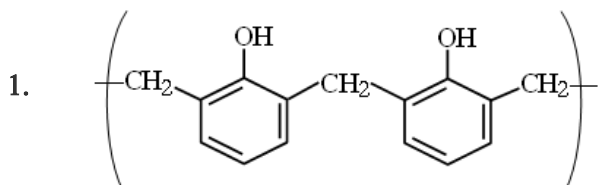
SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

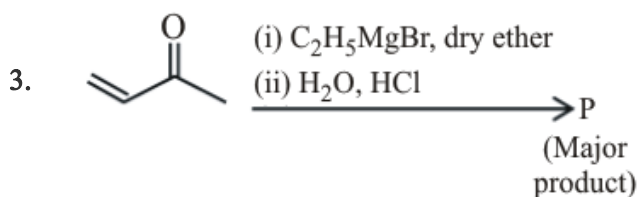


is a repeating unit for :

- (A) Novolac (B) Buna-N
(C) Acrilan (D) Neoprene

2. Which one of the following species responds to an external magnetic field?

- (A) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (B) $[\text{Ni}(\text{CN})_4]^{2-}$
(C) $[\text{Co}(\text{CN})_6]^{3-}$ (D) $[\text{Ni}(\text{CO})_4]$



Consider the above reaction, the major product 'P' is:

- (A)
- (B)
- (C)
- (D)

4. Sodium stearate $\text{CH}_3(\text{CH}_2)_{16}\text{COO}^-\text{Na}^+$ is an anionic surfactant which forms micelles in oil. Choose the **correct** statement for it from the following :

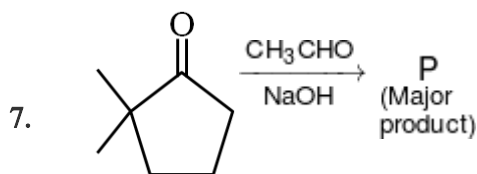
- (A) It forms spherical micelles with $\text{CH}_3(\text{CH}_2)_{16}$ - group pointing towards the centre of sphere.
(B) It forms non-spherical micelles with COO^- group pointing outwards on the surface.
(C) It forms spherical micelles with $\text{CH}_3(\text{CH}_2)_{16}$ - group pointing outwards on the surface of sphere
(D) It forms non-spherical micelles with $\text{CH}_3(\text{CH}_2)_{16}$ - group pointing towards the centre.

5. The water soluble protein is :

- (A) Fibrin
(B) Albumin
(C) Myosin
(D) Collagen

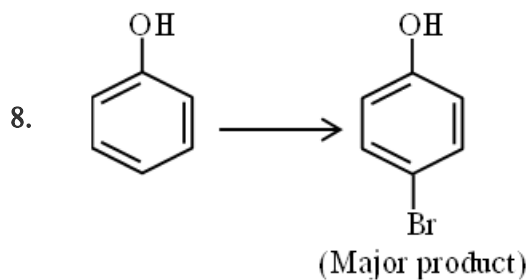
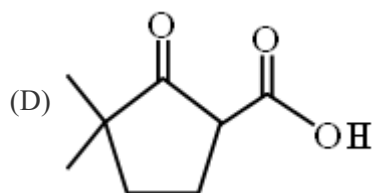
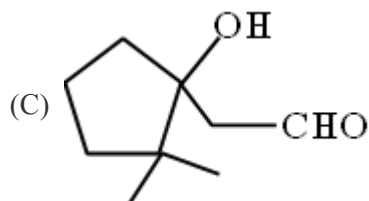
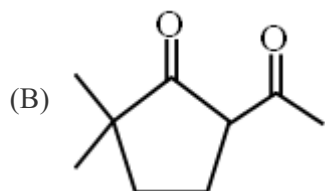
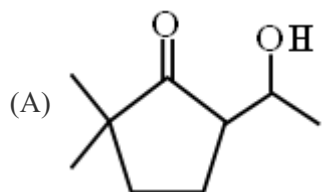
6. At 298.2 K the relationship between enthalpy of bond dissociation (in kJ mol^{-1}) for hydrogen (E_H) and its isotope, deuterium (E_D), is best described by :

- (A) $E_H = \frac{1}{2} E_D$
(B) $E_H = E_D$
(C) $E_H \approx E_D - 7.5$
(D) $E_H = 2E_D$



(i) $I_2/NaOH$, Filter
(ii) Filtrate + $HCl \rightarrow$ 'X'

Consider the given reaction, the product 'X' is:



The given reaction can occur in the presence of :

- (a) Bromine water
- (b) Br_2 in CS_2 , 273 K
- (c) $Br_2/FeBr_3$
- (d) Br_2 in $CHCl_3$, 273 K

Choose the correct answer from the options given below :

- (A) (b) and (d) only
- (B) (a) and (c) only
- (C) (b), (c) and (d) only
- (D) (a), (b) and (d) only

9. Given below are two statements, one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.

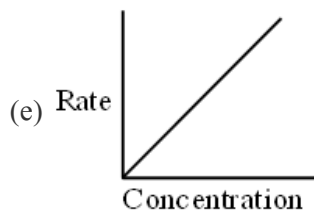
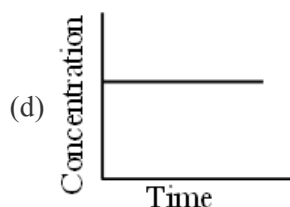
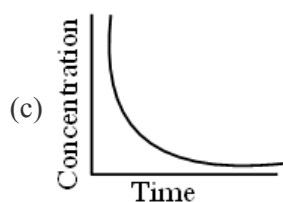
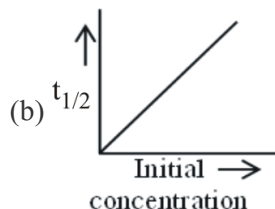
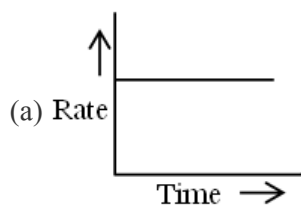
Assertion (R) : Gabriel phthalimide synthesis cannot be used to prepare aromatic primary amines.

Reason : Aryl halides do not undergo nucleophilic substitution reaction.

In the light of the above statements, choose the **correct** answer from the options given below :

- (A) Both **(A)** and **(R)** true but **(R)** is not the correct explanation of **(A)**.
- (B) **(A)** is false but **(R)** is true.
- (C) Both **(A)** and **(R)** true and **(R)** is correct explanation of **(A)**.
- (D) **(A)** is true but **(R)** is false.

10. For the following graphs,

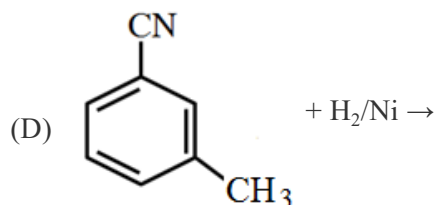
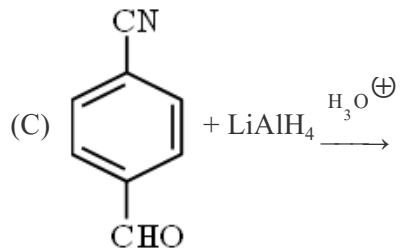
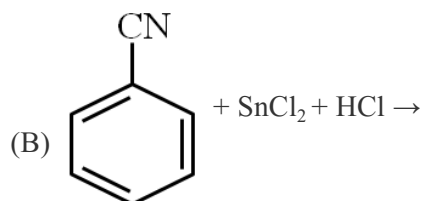
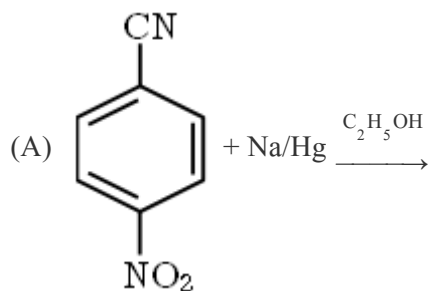


Choose from the options given below, the

correct one regarding order of reaction is :

- (A) (b) zero order (c) and (e) First order
- (B) (a) and (b) Zero order (e) First order
- (C) (b) and (d) Zero order (e) First order
- (D) (a) and (b) Zero order (c) and (e) First order

11. Which one of the products of the following reactions **does not** react with Hinsberg reagent to form sulphonamide?



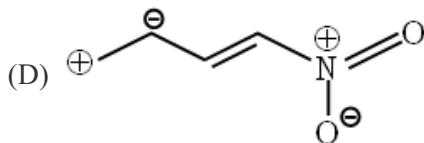
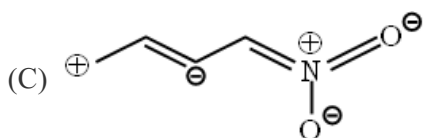
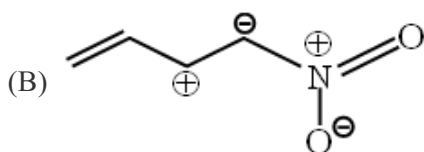
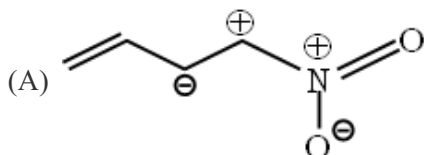
12. The ionic radii of K^+ , Na^+ , Al^{3+} and Mg^{2+} are in the order :

- (A) $\text{Na}^+ < \text{K}^+ < \text{Mg}^{2+} < \text{Al}^{3+}$
- (B) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{K}^+ < \text{Na}^+$
- (C) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{K}^+$
- (D) $\text{K}^+ < \text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+$

13. Which one of the following compounds of Group-14 elements is **not** known?

- (A) $[\text{GeCl}_6]^{2-}$ (B) $[\text{Sn}(\text{OH})_6]^{2-}$
(C) $[\text{SiCl}_6]^{2-}$ (D) $[\text{SiF}_6]^{2-}$

14. Which one among the following resonating structures is **not** correct?



15. Given below are two statements :

Statement I : None of the alkaline earth metal hydroxides dissolve in alkali.

Statement II : Solubility of alkaline earth metal hydroxides in water increases down the group.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (A) **Statement I** is correct but **Statement II** is incorrect.
(B) **Statement I** is incorrect but **Statement II** is correct.
(C) **Statement I** and **Statement II** both are incorrect.
(D) **Statement I** and **Statement II** both are correct.

16. The correct order of following 3d metal oxides, according to their oxidation numbers is :

- (a) CrO_3 (b) Fe_2O_3 (c) MnO_2
(d) V_2O_5 (e) Cu_2O

(A) (d) > (a) > (b) > (c) > (e)

(B) (a) > (c) > (d) > (b) > (e)

(C) (a) > (d) > (c) > (b) > (e)

(D) (c) > (a) > (d) > (e) > (b)

17. Which one of the following chemical agent is **not** being used for dry-cleaning of clothes?

- (A) H_2O_2 (B) CCl_4
(C) Liquid CO_2 (D) $\text{Cl}_2\text{C} = \text{CCl}_2$

18. Which one of the following compounds will liberate CO_2 , when treated with NaHCO_3 ?

- (A) $(\text{CH}_3)_3\text{N}^+\text{HCl}^-$ (B) $(\text{CH}_3)_4\text{N}^+\text{O}^-$

- (C) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$ (D) CH_3NH_2

19. In the leaching of alumina from bauxite, the ore expected to leach out in the process by reacting with NaOH is :

- (A) TiO_2 (B) Fe_2O_3 (C) ZnO (D) SiO_2

20. An organic compound 'A' C_4H_8 on treatment with KMnO_4/H^+ yields compound 'B' $\text{C}_3\text{H}_6\text{O}$. Compound 'A' also yields compound 'B' an ozonolysis. Compound 'A' is :

- (A) 2-Methylpropene
(B) 1-Methylcyclopropane
(C) But-2-ene
(D) Cyclobutane

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type question.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- The number of sigma bonds in $\text{H}_3\text{C} - \underset{\text{H}}{\underset{|}{\text{C}}} = \text{CH} - \text{C} \equiv \text{C} - \text{H}$ is ____.
- Three moles of AgCl get precipitated when one mole of an octahedral co-ordination compound with empirical formula $\text{CrCl}_3 \cdot 3\text{NH}_3 \cdot 3\text{H}_2\text{O}$ reacts with excess of silver nitrate. The number of chloride ions satisfying the secondary valency of the metal ion is ____.
- A source of monochromatic radiation of wavelength 400 nm provides 1000 J of energy in 10 seconds. When this radiation falls on the surface of sodium, $x \times 10^{20}$ electrons are ejected per second. Assume that wavelength 400 nm is sufficient for ejection of electron from the surface of sodium metal. The value of x is _____. (Nearest integer)
($h = 6.626 \times 10^{-34}$ Js)
- CO_2 gas is bubbled through water during a soft drink manufacturing process at 298 K. If CO_2 exerts a partial pressure of 0.835 bar then x m mol of CO_2 would dissolve in 0.9 L of water. The value of x is _____. (Nearest integer)
(Henry's law constant for CO_2 at 298 K is 1.67×10^3 bar)

- For the reaction, $\text{A} + \text{B} \rightleftharpoons 2\text{C}$ the value of equilibrium constant is 100 at 298 K. If the initial concentration of all the three species is 1 M each, then the equilibrium concentration of C is $x \times 10^{-1}$ M. The value of x is _____. (Nearest integer)
- Consider the cell at 25°C
 $\text{Zn} | \text{Zn}^{2+}(\text{aq}), (1 \text{ M}) || \text{Fe}^{3+}(\text{aq}), \text{Fe}^{2+}(\text{aq}) | \text{Pt(s)}$
The fraction of total iron present as Fe^{3+} ion at the cell potential of 1.500 V is $x \times 10^{-2}$. The value of x is _____. (Nearest integer)
(Given : $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 = 0.77\text{V}$, $E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76\text{V}$)
- At 298 K, the enthalpy of fusion of a solid (X) is 2.8 kJ mol^{-1} and the enthalpy of vaporisation of the liquid (X) is 98.2 kJ mol^{-1} . The enthalpy of sublimation of the substance (X) in kJ mol^{-1} is _____. (in nearest integer)
- A home owner uses $4.00 \times 10^3 \text{ m}^3$ of methane (CH_4) gas, (assume CH_4 is an ideal gas) in a year to heat his home. Under the pressure of 1.0 atm and 300 K, mass of gas used is $x \times 10^5$ g. The value of x is _____. (Nearest integer)
(Given $R = 0.083 \text{ L atm K}^{-1} \text{ mol}^{-1}$)
- When 10 mL of an aqueous solution of Fe^{2+} ions was titrated in the presence of dil H_2SO_4 using diphenylamine indicator, 15 mL of 0.02 M solution of $\text{K}_2\text{Cr}_2\text{O}_7$ was required to get the end point. The molarity of the solution containing Fe^{2+} ions is $x \times 10^{-2}$ M. The value of x is _____. (Nearest integer)
- Consider the complete combustion of butane, the amount of butane utilized to produce 72.0 g of water is _____ $\times 10^{-1}$ g. (in nearest integer)

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. A spherical gas balloon of radius 16 meter subtends an angle 60° at the eye of the observer A while the angle of elevation of its center from the eye of A is 75° . Then the height (in meter) of the top most point of the balloon from the level of the observer's eye is :

- (A) $8(2 + 2\sqrt{3} + \sqrt{2})$
 (B) $8(\sqrt{6} + \sqrt{2} + 2)$
 (C) $8(\sqrt{2} + 2 + \sqrt{3})$
 (D) $8(\sqrt{6} - \sqrt{2} + 2)$

2. Let $f(x) = 3\sin^4 x + 10\sin^3 x + 6\sin^2 x - 3$, $x \in \left[-\frac{\pi}{6}, \frac{\pi}{2}\right]$. Then, f is :

- (A) increasing in $\left(-\frac{\pi}{6}, \frac{\pi}{2}\right)$
 (B) decreasing in $\left(0, \frac{\pi}{2}\right)$
 (C) increasing in $\left(-\frac{\pi}{6}, 0\right)$
 (D) decreasing in $\left(-\frac{\pi}{6}, 0\right)$

3. Let S_n be the sum of the first n terms of an arithmetic progression. If $S_{3n} = 3S_{2n}$, then the value of $\frac{S_{4n}}{S_{2n}}$ is :

- (A) 6 (B) 4 (C) 2 (D) 8

4. The locus of the centroid of the triangle formed by any point P on the hyperbola

$$16x^2 - 9y^2 + 32x + 36y - 164 = 0, \text{ and its foci is :}$$

- (A) $16x^2 - 9y^2 + 32x + 36y - 36 = 0$
 (B) $9x^2 - 16y^2 + 36x + 32y - 144 = 0$
 (C) $16x^2 - 9y^2 + 32x + 36y - 144 = 0$
 (D) $9x^2 - 16y^2 + 36x + 32y - 36 = 0$

5. Let the vectors

$$(2 + a + b)\hat{i} + (a + 2b + c)\hat{j} - (b + c)\hat{k},$$

$$(1 + b)\hat{i} + 2b\hat{j} - b\hat{k} \text{ and}$$

$$(2 + b)\hat{i} + 2b\hat{j} + (1 - b)\hat{k} \text{ a, b, c, } \in \mathbf{R} \text{ be}$$

co-planar. Then which of the following is true?

- (A) $2b = a + c$ (B) $3c = a + b$
 (C) $a = b + 2c$ (D) $2a = b + c$

6. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be defined as

$$f(x) = \begin{cases} \frac{\lambda |x^2 - 5x + 6|}{\mu(5x - x^2 - 6)}, & x < 2 \\ e^{\frac{\tan(x-2)}{x-[x]}} & , x > 2 \\ \mu & , x = 2 \end{cases}$$

where $[x]$ is the greatest integer less than or equal to x. If f is continuous at $x = 2$, then $\lambda + \mu$ is equal to :

- (A) $e(-e + 1)$ (B) $e(e - 2)$
 (C) 1 (D) $2e - 1$

7. The value of the definite integral

$$\int_{\pi/24}^{5\pi/24} \frac{dx}{1 + \sqrt[3]{\tan 2x}} \text{ is :}$$

- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$
(C) $\frac{\pi}{12}$ (D) $\frac{\pi}{18}$

8. If b is very small as compared to the value of a , so that the cube and other higher powers of $\frac{b}{a}$ can be neglected in the identity

$$\frac{1}{a-b} + \frac{1}{a-2b} + \frac{1}{a-3b} + \dots + \frac{1}{a-nb} = \alpha n + \beta n^2 + \gamma n^3, \text{ then the value of } \gamma \text{ is :}$$

- (A) $\frac{a^2 + b}{3a^3}$ (B) $\frac{a + b}{3a^2}$
(C) $\frac{b^2}{3a^3}$ (D) $\frac{a + b^2}{3a^3}$

9. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = 1 + x e^{y-x}, -\sqrt{2} < x < \sqrt{2}, y(0) = 0$ then, the minimum value of $y(x)$, $x \in (-\sqrt{2}, \sqrt{2})$ is equal to :

- (A) $(2 - \sqrt{3}) - \log_e 2$
(B) $(2 + \sqrt{3}) + \log_e 2$
(C) $(1 + \sqrt{3}) - \log_e (\sqrt{3} - 1)$
(D) $(1 - \sqrt{3}) - \log_e (\sqrt{3} - 1)$

10. The Boolean expression $(p \Rightarrow q) \wedge (q \Rightarrow \sim p)$ is equivalent to :

- (A) $\sim q$ (B) q
(C) p (D) $\sim p$

11. The area (in sq. units) of the region, given by the set $\{(x, y) \in \mathbf{R} \times \mathbf{R} \mid x \geq 0, 2x^2 \leq y \leq 4 - 2x\}$ is :

- (A) $\frac{8}{3}$ (B) $\frac{17}{3}$ (C) $\frac{13}{3}$ (D) $\frac{7}{3}$

12. The sum of all values of x in $[0, 2\pi]$, for which $\sin x + \sin 2x + \sin 3x + \sin 4x = 0$, is equal to :

- (A) 8π (B) 11π (C) 12π (D) 9π

13. Let $g : \mathbf{N} \rightarrow \mathbf{N}$ be defined as

$$g(3n + 1) = 3n + 2,$$

$$g(3n + 2) = 3n + 3,$$

$$g(3n + 3) = 3n + 1, \text{ for all } n \geq 0.$$

Then which of the following statements is true ?

- (A) There exists an onto function $f : \mathbf{N} \rightarrow \mathbf{N}$ such that $fog = f$
(B) There exists a one-one function $f : \mathbf{N} \rightarrow \mathbf{N}$ such that $fog = f$
(C) $gogog = g$
(D) There exists a function $f : \mathbf{N} \rightarrow \mathbf{N}$ such that $gof = f$

14. Let $f : [0, \infty) \rightarrow [0, \infty)$ be defined as

$$f(x) = \int_0^x [y] dy$$

where $[x]$ is the greatest integer less than or equal to x . Which of the following is true?

- (A) f is continuous at every point in $[0, \infty)$ and differentiable except at the integer points.
(B) f is both continuous and differentiable except at the integer points in $[0, \infty)$.
(C) f is continuous everywhere except at the integer points in $[0, \infty)$.
(D) f is differentiable at every point in $[0, \infty)$.

15. The values of a and b , for which the system of equations
 $2x + 3y + 6z = 8$
 $x + 2y + az = 5$
 $3x + 5y + 9z = b$
 has no solution, are :
 (A) $a = 3, b \neq 13$
 (B) $a \neq 3, b \neq 13$
 (C) $a \neq 3, b = 13$
 (D) $a = 3, b = 13$
16. Let 9 distinct balls be distributed among 4 boxes, B_1, B_2, B_3 and B_4 . If the probability that B_3 contains exactly 3 balls is $k\left(\frac{3}{4}\right)^9$ then k lies in the set :
 (A) $\{x \in \mathbf{R} : |x - 3| < 1\}$
 (B) $\{x \in \mathbf{R} : |x - 2| \leq 1\}$
 (C) $\{x \in \mathbf{R} : |x - 1| < 1\}$
 (D) $\{x \in \mathbf{R} : |x - 5| \leq 1\}$
17. Let a parabola P be such that its vertex and focus lie on the positive x -axis at a distance 2 and 4 units from the origin, respectively. If tangents are drawn from $O(0, 0)$ to the parabola P which meet P at S and R , then the area (in sq. units) of ΔSOR is equal to :
 (A) $16\sqrt{2}$
 (B) 16
 (C) 32
 (D) $8\sqrt{2}$
18. The number of real roots of the equation
 $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$ is :
 (A) 2 (B) 4
 (C) 6 (D) 1
19. Let an ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$, passes through $\left(\sqrt{\frac{3}{2}}, 1\right)$ and has eccentricity $\frac{1}{\sqrt{3}}$. If a circle, centered at focus $F(a, 0), a > 0$, of E and radius $\frac{2}{\sqrt{3}}$, intersects E at two points P and Q , then PQ^2 is equal to :
 (A) $\frac{8}{3}$
 (B) $\frac{4}{3}$
 (C) $\frac{16}{3}$
 (D) 3
20. Let the foot of perpendicular from a point $P(1, 2, -1)$ to the straight line $L : \frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$ be N . Let a line be drawn from P parallel to the plane $x + y + 2z = 0$ which meets L at point Q . If α is the acute angle between the lines PN and PQ , then $\cos \alpha$ is equal to _____.
 (A) $\frac{1}{\sqrt{5}}$ (B) $\frac{\sqrt{3}}{2}$
 (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2\sqrt{3}}$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Let $y = y(x)$ be solution of the following differential equation $e^y \frac{dy}{dx} - 2e^y \sin x + \sin x \cos^2 x = 0$, $y\left(\frac{\pi}{2}\right) = 0$. If $y(0) = \log_e(\alpha + \beta e^{-2})$, then $4(\alpha + \beta)$ is equal to _____.

2. If the value of $\left(1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \dots \text{upto } \infty\right)^{\log_{(0.25)}\left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots \text{upto } \infty\right)}$ is ℓ , then ℓ^2 is equal to _____.

3. Consider the following frequency distribution :

Class :	10-20	20-30	30-40	40-50	50-60
Frequency :	α	110	54	30	β

If the sum of all frequencies is 584 and median is 45, then $|\alpha - \beta|$ is equal to _____.

4. Let $\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. If a vector $\vec{r} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ is perpendicular to each of the vectors $(\vec{p} + \vec{q})$ and $(\vec{p} - \vec{q})$, and $|\vec{r}| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to _____.

5. The ratio of the coefficient of the middle term in the expansion of $(1 + x)^{20}$ and the sum of the coefficients of two middle terms in expansion of $(1 + x)^{19}$ is _____.

6. Let $M = \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \{\pm 3, \pm 2, \pm 1, 0\} \right\}$.

Define $f : M \rightarrow \mathbf{Z}$, as $f(A) = \det(A)$, for all $A \in M$, where $\mathbf{Z}$ is set of all integers. Then the number of $A \in M$ such that $f(A) = 15$ is equal to _____.

7. There are 5 students in class 10, 6 students in class 11 and 8 students in class 12. If the number of ways, in which 10 students can be selected from them so as to include at least 2 students from each class and at most 5 students from the total 11 students of class 10 and 11 is $100k$, then k is equal to _____.

8. If α, β are roots of the equation $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ for each positive integer n , then the value of $\left(\frac{P_{17}P_{20} + 5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19} + 5\sqrt{2}P_{18}^2} \right)$ is equal to _____.

9. The term independent of 'x' in the expansion of $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}} \right)^{10}$, where $x \neq 0, 1$ is equal to _____.

10. Let $S = \left\{ n \in \mathbf{N} \mid \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \forall a, b, c, d \in \mathbf{R} \right\}$, where $i = \sqrt{-1}$. Then the number of 2-digit numbers in the set S is _____.

Space for Rough Work

